TRAVELEASE: AI-Powered Travel Made Easy

Manav Prajith M, Devika K, Aswini R, Abhishek Sreekumar, Parvathy E N
Department of AI& DS
Muthoot Institute of Technology and Science
Kochi, Kerala, India
22ad450@mgits.ac.in, 22ad455@mgits.ac.in, 22ad457@mgits.ac.in, 22ad458@mgits.ac.in, priyacv@mgits.ac.in

***Abstract*—TravelEase is an AI-driven itinerary generation system that creates personalized, day-wise travel plans by combining multimodal user inputs with intelligent data processing. As highlighted by Rajput et al. [1], traditional travel recommendation systems often provide isolated suggestions rather than cohesive itineraries, creating a significant gap between raw travel information and meaningful travel experiences. The complexity of modern travel decision-making, shaped by cost fluctuations and diverse preferences, demands solutions capable of synthesizing multiple factors simultaneously [2, 3]. TravelEase addresses this gap by allowing users to provide structured information such as destination, duration, and budget, along with an inspiration image that reflects their preferred travel style. The image is analyzed using EfficientNet, an advanced convolutional neural network architecture that efficiently extracts high-level visual features and contextual patterns as demonstrated by Zhang et al. [4] for fine-grained visual recognition tasks. The system builds upon the data-driven approach established by Shrestha et al. [5] for building comprehensive tourism databases, and follows the evaluation methodologies proposed by Veluru [6] for assessing AI-enhanced travel planning systems. Spatial clustering techniques are then applied to group nearby attractions into geographically coherent daily schedules, reducing travel distance and improving route efficiency [7]. The system follows a modular microservices architecture consisting of a React-based frontend, a Node.js backend for geospatial processing, and a Flask-based vision service for EfficientNet inference, ensuring scalability and maintainability [8]. Intelligent fallback mechanisms ensure reliable itinerary generation even when external APIs fail, making TravelEase a scalable and adaptive solution for smart travel planning [9].**

***Index Terms*—EfficientNet, Convolutional Neural Networks (CNN), Image Feature Extraction, Multimodal Input Processing, Geospatial Data Integration, OpenStreetMap (OSM), Overpass API, Spatial Clustering, Itinerary Optimization, Microservices Architecture, React, Node.js, Flask, Supabase, Personalized Recommendation Systems, Intelligent Fallback Mechanisms**

I. INTRODUCTION

The rapid evolution of digital tourism platforms has demonstrated the pressing need for automated systems that can transform fragmented travel information into cohesive and highly personalized itineraries [1]. Buhalis and Law [10] noted that progress in information technology and tourism management has fundamentally transformed the eTourism landscape over two decades. As observed by Veluru [6], the travel industry has undergone a revolution due to rapid technological advancements, with Generative AI and Artificial Intelligence emerging as game-changers in providing best-in-class solutions for travel planning processes. Gretzel [11] emphasized that intelligent systems in tourism require a social science perspective to truly understand traveler needs and behaviors.

TravelEase is an AI-powered smart travel planning system designed to generate personalized travel itineraries by combining deep learning-based image analysis with real-world geographic data. The difficulty in travel planning lies not in accessing information, but in organizing it in a way that aligns with individual travel styles [1]. A planned trip must respect personal preferences, daily energy levels, destination characteristics, and logistics such as transportation and opening hours [5]. Travelers now seek personalized experiences rather than generic lists of sites, valuing plans that reflect their interests, whether cultural, recreational, adventurous, historical, or nature-focused [12].

The system leverages EfficientNet, a state-of-the-art convolutional neural network architecture known for its optimized accuracy and computational efficiency. Lecun et al. [13] established that deep learning techniques have achieved excellent performance in many image processing tasks, especially surpassing human recognition accuracy in image classification tasks. Zhang et al. [4] demonstrated that EfficientNet architectures employ compound scaling to balance network depth, width, and resolution, achieving state-of-the-art accuracy with fewer parameters than conventional models. Tan and Le [14] introduced the compound scaling method that uniformly scales all dimensions of network depth, width, and resolution using a simple yet effective coefficient.

To ensure realistic and location-based recommendations, TravelEase integrates OpenStreetMap data through the Overpass API to fetch real-world tourist attractions, museums, historic sites, and other points of interest. Shrestha et al. [5] developed a comprehensive database for Pokhara City, Nepal, using information from various sources including the Nepal Tourism Board, TripAdvisor, Google search, and travel websites. Devkota et al. [15] demonstrated the effectiveness of using volunteered geographic information to identify tourism areas of interest. The integration of geospatial data with user preferences follows the framework proposed by Alrasheed et al. [16] for multi-level tourism destination recommendation.

By combining deep learning-based visual intelligence with geospatial data processing, TravelEase delivers a scalable, adaptive, and intelligent travel experience that goes beyond conventional itinerary generation systems, addressing the critical need identified by Ricci et al. [17] for recommender systems that can handle the unique challenges of the tourism domain.

II. Related Works

A. Evolution of Travel Recommendation Systems

Early research in e-tourism was dominated by two primary filtering paradigms: Collaborative Filtering (CF) and Content-Based Filtering (CBF) [18]. As documented by Rajput et al. [1], CF algorithms, such as Matrix Factorization and Nearest Neighbor, operate on the assumption that users who agreed in the past will agree in the future. Gavalas et al. [2] provided a comprehensive review of mobile recommender systems in tourism, highlighting both the potential and limitations of traditional approaches.

However, CF suffers significantly from the "Data Sparsity" problem in the tourism domain, where users visit only a fraction of available destinations, leading to sparse user-item interaction matrices that degrade recommendation quality [1]. Furthermore, CF is plagued by the "Cold Start" problem, where the system fails to provide relevant suggestions for new users with no prior booking history [8]. Schafer et al. [19] discussed these challenges in the context of e-commerce recommender systems.

Content-Based Filtering approaches attempt to solve this by matching user profiles with item attributes [5]. Users expressing interest in "history" receive recommendations for sites tagged "monument." However, these systems are often limited by "Over-Specialization," where the user is restricted to seeing items similar to what they have already interacted with, preventing the discovery of novel experiences [1]. Burke [20] surveyed hybrid recommender systems that combine multiple approaches to overcome individual limitations.

Crucially, as Rajput et al. [1] note, neither CF nor CBF can inherently generate a sequential itinerary; they provide a list of isolated spots without considering logistical constraints like travel time between locations or opening hours. Wang et al. [21] highlighted the role of smartphones in mediating the touristic experience, suggesting the need for more intelligent, context-aware systems.

B. Deep Learning for Point-of-Interest Prediction

To address the sequential nature of travel, researchers introduced Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for next Point-of-Interest (POI) prediction [1]. Studies have demonstrated that LSTMs can effectively analyze a tourist's historical trajectory to predict their next likely stop [22]. Zhang et al. [23] provided a comprehensive survey of deep learning based recommender systems, highlighting the evolution from traditional methods to neural approaches.

Nilashi et al. [24] employed a fuzzy-logic approach for analyzing travelers' decision-making using online reviews in social network sites. Arreeras et al. [25] used association rule mining to identify tourist-attractive destinations for sustainable development of tourism areas using Wi-Fi tracking data. Essien and Chukwukelu [26] discussed deep learning applications in hospitality and tourism, proposing a research framework agenda for future work.

Despite improved performance, predictive models operate as discriminative black boxes lacking explainability [1]. They cannot synthesize novel itineraries from abstract constraints such as "plan a budget-friendly cultural trip for an elderly couple." These models require massive structured datasets of user trajectories, which are often unavailable due to privacy restrictions or collection costs [5].

C. Generative AI and Large Language Models

The emergence of Transformer-based architectures has marked a paradigm shift from "Information Retrieval" to "Information Synthesis" [1]. Vaswani et al. [27] introduced the self-attention mechanism that captures long-range dependencies, forming the foundation of modern Large Language Models. Large Language Models (LLMs) such as GPT-4 and Google Gemini utilize massive training corpora to perform "Zero-Shot" and "Few-Shot" reasoning about complex queries [9].

Veluru [6] highlights that Generative AI can provide highly personalized trip itineraries and travel content according to user preferences, improving the experience of planning to travel while saving much time and effort. Li et al. [28] provided a comprehensive survey on Large Language Models for recommendation, discussing how LLMs can process natural language constraints that are difficult to encode mathematically. Liu et al. [29] explored personalized recommendation via prompting Large Language Models.

Recent work explores LLMs as autonomous agents capable of decomposing complex goals into sub-tasks [1]. However, raw LLM outputs suffer from hallucination—inventing nonexistent places—necessitating structured application layers for production deployment [6]. OpenAI [30] documented the capabilities and limitations of GPT-4, providing insights into responsible deployment of large language models.

D. Computer Vision in Tourism

Visual content plays a crucial role in travel decision-making. Zhang et al. [4] demonstrated that deep learning techniques represented by convolutional neural networks have achieved excellent performance in many image processing tasks, especially surpassing human recognition accuracy in image classification tasks. Their work on fine-grained car recognition using lightweight attention networks showed that adding appropriate attention mechanism modules to capture and focus on key positions in classification networks is an effective solution for identifying subtle differences.

Hu et al. [31] introduced the Squeeze-and-Excitation (SE) block, which adaptively recalibrates channel-wise feature responses by explicitly modeling interdependencies between channels. Woo et al. [32] proposed the Convolutional Block Attention Module (CBAM), a simple yet effective attention module for feed-forward convolutional neural networks. Yang et al. [33] developed SimAM, a simple, parameter-free attention module for convolutional neural networks.

EfficientNet architectures, as employed by Zhang et al. [4], achieve superior performance with high cost-effectiveness.

Iandola et al. [34] developed SqueezeNet, achieving AlexNet-level accuracy with 50x fewer parameters. Howard et al. [35] introduced MobileNets, efficient convolutional neural networks for mobile vision applications. Sandler et al. [36] proposed MobileNetV2 with inverted residuals and linear bottlenecks. Howard et al. [37] presented MobileNetV3, searching for optimal network architecture. Ma et al. [38] provided practical guidelines for efficient CNN architecture design with ShuffleNet V2.

E. Data-Driven Tourism Recommendation Systems

Shrestha et al. [5] conducted an extensive study on tourist planning assessment, spending nature, preference indicators, and satisfaction, collecting data from 2400 international and domestic tourists through structured questionnaires. Their work established that a robust Tourist Recommender System must include attributes such as demographics, destination-planning traits, behavior, spending tendencies, preferences, and satisfaction metrics.

Kulshrestha et al. [39] proposed a Bayesian BILSTM approach for tourism demand forecasting. Hu et al. [40] developed hierarchical pattern recognition for tourism demand forecasting. Pai and Hong [41] used support vector machines for forecasting tourism demand with multifactor models. Maravanyika and Dlodlo [42] proposed an adaptive framework for recommender-based learning management systems.

Their comparative analysis revealed that existing recommendation systems from TripAdvisor, Google Maps, and tourism portals are generic, offer static information, lack real-time updates, and suffer from accuracy and precision problems [5]. In contrast, their personalized recommender system using supervised machine learning algorithms (Random Forest, Gradient Boost, Decision Tree) achieved high accuracy, with Random Forest reaching 94% accuracy for planning models and Gradient Boost achieving 100% for behavioral models.

F. Itinerary Optimization and Route Planning

Veluru [6] notes that AI has made immense contributions to route optimization, with AI-based planners optimizing efficient travel routes by considering factors like traffic, distance, user preferences, and possible delays. Chen et al. [7] developed AI-based route optimization for travel planning, demonstrating significant improvements in travel efficiency.

Rajput et al. [1] emphasize that the core of effective itinerary generation lies in geographical clustering, time allocation and duration estimation, activity balancing, logical morning-evening segmentation, and incorporation of breaks, meals, and buffer time. Their proposed algorithm follows a "Generate-then-Validate" approach, strictly enforcing input feasibility before AI inference and employing schema validation to ensure output quality.

Kbaier et al. [43] developed a personalized hybrid tourist recommender system using different machine-learning algorithms including K-NN for both collaborative and content-based filtering and decision trees for the decision fusion. Shambour [44] proposed a deep learning-based algorithm for multi-criteria recommender systems. Yassine et al. [45] developed an intelligent recommender system based on unsupervised machine learning and demographic attributes.

G. Research Gap

Despite extensive work on recommendation algorithms, POI prediction, and route optimization, several gaps remain. Rajput et al. [1] note that although there is abundant work on generating recommendation lists, relatively fewer studies offer solutions that produce structured itineraries combining recommendation accuracy with day-wise organization, contextual awareness, and real-world feasibility. Shrestha et al. [5] identify that research in model-based filtering using a combination of tourist attributes (planning, behavioral, preferences, and satisfaction), social data, and machine learning is not available, particularly in local contexts. Gössling [46] discussed the role of technology and ICT in tourism from big data to the big picture, highlighting the need for integrated approaches.

Wenan et al. [47] performed analysis and design of tourist recommender systems for religious destinations in Nepal. Shrestha et al. [48] conducted preliminary analysis and design of a customized tourist recommender system. Devkota and Miyazaki [49] performed an exploratory study on the generation and distribution of geotagged tweets in Nepal. Devkota et al. [50] utilized user-generated contents to describe tourism areas of interest. TravelEase addresses these gaps through a multimodal architecture combining computer vision, geospatial intelligence, and optimization algorithms.

III. Proposed System

The proposed system, TravelEase, is an AI-driven intelligent travel planning platform designed to generate personalized and optimized itineraries using multimodal inputs. As advocated by Rajput et al. [1], the methodology prioritizes organization, clarity, and sequencing, ensuring that travelers receive itineraries that not only highlight important attractions but also provide a balanced distribution of activities across the entire trip.

The system leverages a CNN-based vision module powered by EfficientNet to analyze user-uploaded travel images and extract meaningful semantic features. Zhang et al. [4] demonstrated that lightweight attention networks with hybrid attention modules can effectively focus on regions consistent with target classes while suppressing background and task-irrelevant noise. Their visualization results using Grad-CAM showed that networks with proper attention mechanisms significantly focus and reorganize attention areas, suppressing contribution of irrelevant background areas [51].

Following the methodology of Shrestha et al. [5], TravelEase collects and processes user data including destination, duration, budget, and preferences. The system employs data preprocessing techniques including data integration, cleaning, reduction, and transformation, with imputation methods applied to missing data. Outlier detection is performed using the

Interquartile Range (IQR) method as described in their work [5].

The vision-based preference extraction module classifies user preferences into categories such as natural landscapes, urban environments, heritage sites, or scenic locations. This multi-label classification approach draws inspiration from the fine-grained classification methods discussed by Wei et al. [52], who surveyed deep learning approaches for fine-grained image analysis. The system extracts semantic embeddings and generates travel vibe tags to guide POI selection, similar to how Zheng et al. [53] applied AI for personalized recommendations in travel.

For POI retrieval and ranking, TravelEase integrates geospatial data from OpenStreetMap through the Overpass API. Badaro et al. [54] proposed hybrid approaches with collaborative filtering for recommender systems, which informs the ranking methodology. The system evaluates retrieved POIs based on relevance score, proximity, category match, and optional budget filters, following the parametric weighted approach where weights are determined based on survey data analysis, user feedback, and expert knowledge [5].

The spatial clustering and route optimization module organizes selected POIs into day-wise itineraries using algorithms such as K-Means and DBSCAN. Lee et al. [55] designed and implemented a tour planning system for telematics users, providing foundational work for route optimization in tour

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